ANALYSIS II — SHEET 4 (for Week 5 tutorials)
CIFT. Uniform Continuity.

1. For each of the following functions $f_i\colon (-1,1) \to \mathbb{R}$, determine the image $S_i = f_i((-1,1))$ and state whether or not

(i) S_i is an interval,

(ii) f_i has an inverse defined on S_i,

(iii) f_i has a continuous inverse defined on S_i.

$$f_1(x) = x^2, \qquad f_2(x) = \begin{cases} x & -1 < x \leqslant 0, \\ x+1 & 0 < x < 1, \end{cases} \qquad f_3(x) = \frac{x}{1-|x|}.$$

2. The function $f\colon [0,1] \to [0,1]$ is such that $f(f(x)) = x$ for all $x \in [0,1]$.

(i) Show that f is a bijection.

(ii) Show that if f is strictly increasing then $f(x) = x$ for all $x \in [0,1]$.

(iii) Show that there are infinitely many such f which are strictly decreasing.

(iv) Give an example of such an f which is discontinuous.

(v) Find a continuous, strictly increasing function $g\colon [0,1] \to [0,1]$ such that $g(g(x)) = x^2$ for all $x \in [0,1]$. (This g is not unique; see Exercise 10.)

(vi) Show that there is no continuous function $h\colon [0,\pi/2] \to [0,\pi/2]$ such that $h(h(x)) = \cos x$ for all $x \in [0,\pi/2]$.

3. (i) For each $\lambda \geqslant e$, show that there is a unique $x \in (0,1]$ satisfying $e^x = \lambda x$.

(ii) Show further that the map that takes λ to x is a continuous function on $[e,\infty)$.

[Hint: $e^x \geqslant 1 + x$ for all $x \in \mathbb{R}$.]

4. Which of the following functions are uniformly continuous on $(0,\infty)$? Are any Lipschitz continuous? Justify your answers.

$$\text{(i)} \quad e^{-x}, \qquad \text{(ii)} \quad \sqrt{x}, \qquad \text{(iii)} \quad \frac{1}{\sqrt{x}}, \qquad \text{(iv)} \quad \sin(e^x).$$

5. Suppose that $f\colon \mathbb{R} \to \mathbb{R}$ is uniformly continuous. Show that $f(x+1) - f(x)$ is bounded.

6. (i) Suppose that $f\colon (a,b] \to \mathbb{R}$ is uniformly continuous and that f converges to a limit as $x \to a$. Show that f is uniformly continuous on $(a,b]$.

(ii) Suppose now that $g\colon (a,b] \to \mathbb{R}$ is uniformly continuous and (p_n) is a Cauchy sequence in $(a,b]$. Show that $(g(p_n))$ is also Cauchy.

(iii) Deduce that if $g\colon (a,b] \to \mathbb{R}$ is uniformly continuous then $g(x)$ converges to a limit as $x \to a$.

Extension Exercises (Optional)

7. Let $f, g\colon \mathbb{R} \to \mathbb{R}$ be uniformly continuous. Which of the following functions are necessarily uniformly continuous? Give a proof or counterexample.

(i) $f(x)\, g(x)$,

(ii) $f(x) + g(x)$,

(iii) $f(g(x))$.

8. Suppose that $f\colon \mathbb{R} \to \mathbb{R}$ is continuous and $f(x+1) - f(x)$ is bounded. Need f be uniformly continuous?

9. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a function with the property that the pre-image of any open interval is an open interval. (Note an open interval may be empty or unbounded.)

(i) Give an example of a monotone function which does not have this property.
(ii) Give an example of a continuous function which does not have this property.
(iii) Show that a monotone, continuous function has the given property.
(iv) Show that a function with the given property is necessarily continuous.
(v) Show that a function with the given property is necessarily monotone.

10. The function $g\colon [0,1] \to [0,1]$ is constructed as follows. Take $x_0 \in (0,1)$ and define

$$x_n = (x_0)^{2^{n/2}} \in (0,1) \quad \text{for } n \in \mathbb{Z}.$$

Let g be any continuous, strictly increasing function which maps $[x_1, x_0]$ onto $[x_2, x_1]$. Extend g to $[x_2, x_1]$ by defining $g(x) = \left(g^{-1}(x)\right)^2$. Then inductively extend g to $[x_{2k}, x_{2k-2}]$ by

$$\begin{aligned} g(x) &= g\left(\sqrt{x}\right)^2 \qquad \text{for } k > 1; \\ g(x) &= \sqrt{g(x^2)} \qquad \text{for } k \leqslant 0. \end{aligned}$$

Finally we define $g(0) = 0$ and $g(1) = 1$.

Show that $g\colon [0,1] \to [0,1]$ is well-defined, continuous, strictly increasing and satisfies

$$g(g(x)) = x^2 \qquad \text{for all } x \in [0,1].$$

11. It follows from Sheet 3, Exercise 12, that if $f, g\colon [a,b] \to \mathbb{R}$ are increasing then $f - g$ has bounded variation. We prove the converse here.

(i) Let $f\colon [a,b] \to \mathbb{R}$ be a function of bounded variation. For $a < c < b$ show that

$$V_{[a,b]}(f) = V_{[a,c]}(f) + V_{[c,b]}(f).$$

(ii) Show that $V_{[a,x]}(f)$ is an increasing function of x on $[a,b]$.
(iii) Show that $V_{[a,x]}(f) - f(x)$ is an increasing function of x on $[a,b]$.
(iv) Deduce that f can be written as the difference of two increasing functions.